ICAC 2026 · International Conference on Automation and Computing

A Lightweight LLM-Driven Multi-Agent Framework for Virtual Society Simulation with Memory Feedback

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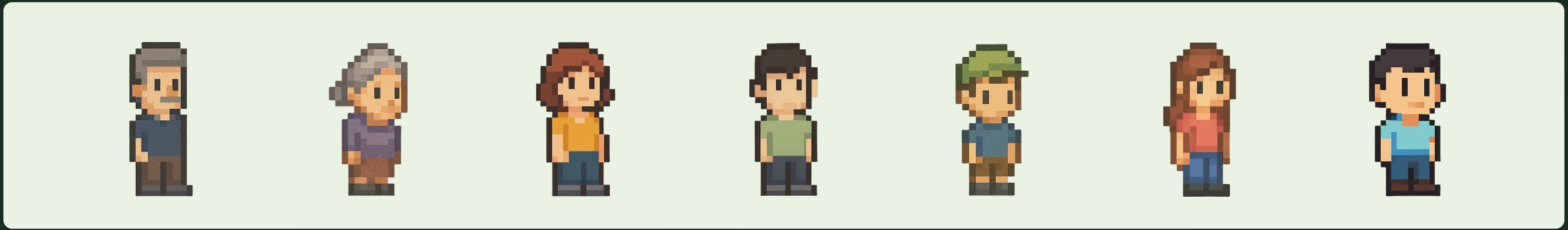
Paper 108 · Parallel Session 3: Robotics & Autonomous Systems · Thursday 27 August 2026



Valentown — seven residents who plan, meet, remember and adapt

MOTIVATION

Two ways to simulate people

R Rule-based agent models

Fixed rules → **Action**

same input → the same behaviour, every day

No context · no planning · no memory

Macro patterns yes — human-like behaviour no

L LLM-driven agents

Prompt → **LLM** → **Action**

fluent — but one step at a time

Reactive: answers the prompt in front of it

Interaction ad hoc or random · no day structure

Missing: explicit structure for behaviour, interaction and feedback

PROBLEM STATEMENT

One question, three requirements

How can we design a virtual society whose agents are structured, evolving and controllable over time?

1 Temporal structure

A day — not a stream of reactions

OUR ANSWER

Schedule-driven agents

2 Behavioural evolution

Yesterday changes tomorrow

OUR ANSWER

Memory feedback loop

3 Controllability

An explicit rule, plus a human override

OUR ANSWER

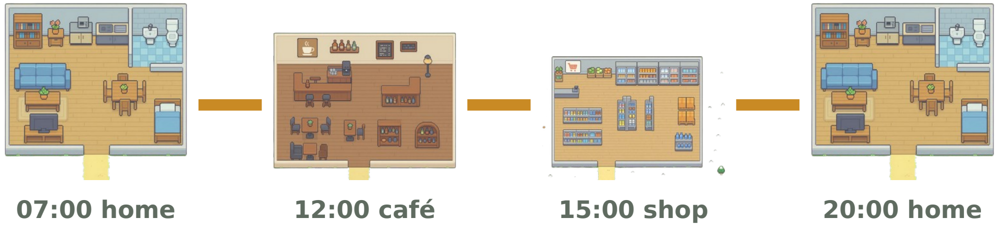
Co-location trigger + human-in-the-loop

MECHANISM 1 & 2

Plan the day, then let the town decide who meets

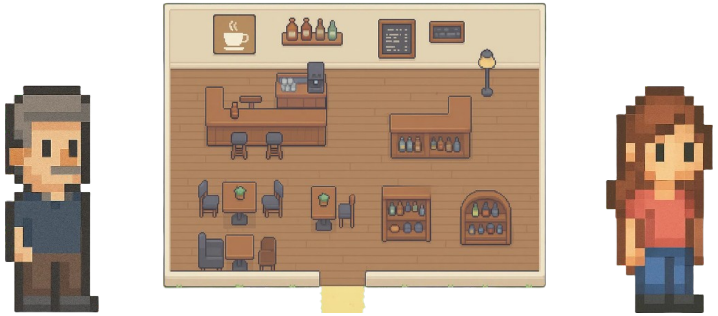
1 Schedule-driven day

identity + goals + memory → one day's plan



one generated day, illustrative

2 Co-location interaction



same place + same period → they talk

Deterministic when · generative what

the framework controls the dynamics — the model supplies the diversity

MECHANISM 3

Memory is where the days connect

3 Memory-based feedback loop

WRITTEN IN

Plans · Conversations · Reflections

READ OUT

Planning · Dialogue · Reflection

Every entry: category · timestamp · importance score

Retrieval: recent + important · the store stays bounded

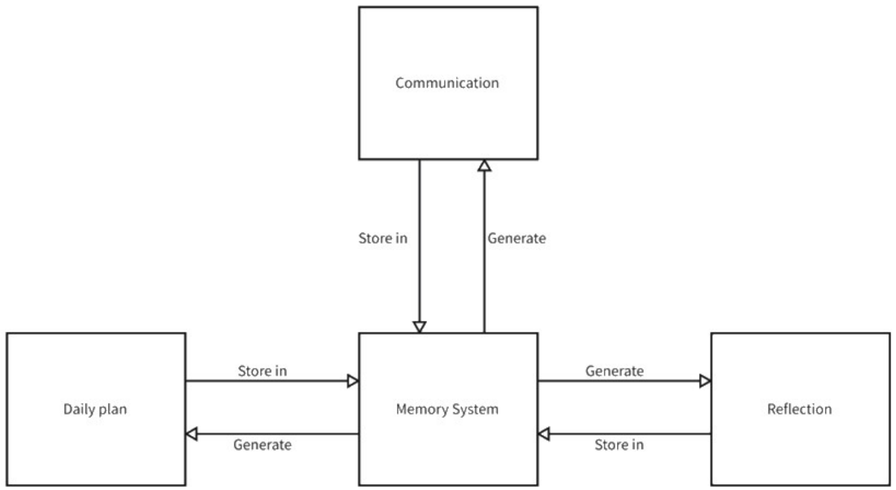


Fig. 3 Memory-centred feedback loop

OPTIONAL: HUMAN IN THE LOOP

Feedback written into the same memory channel

The LLM stays stateless — all continuity lives in memory

EXECUTION

One simulation day, end to end

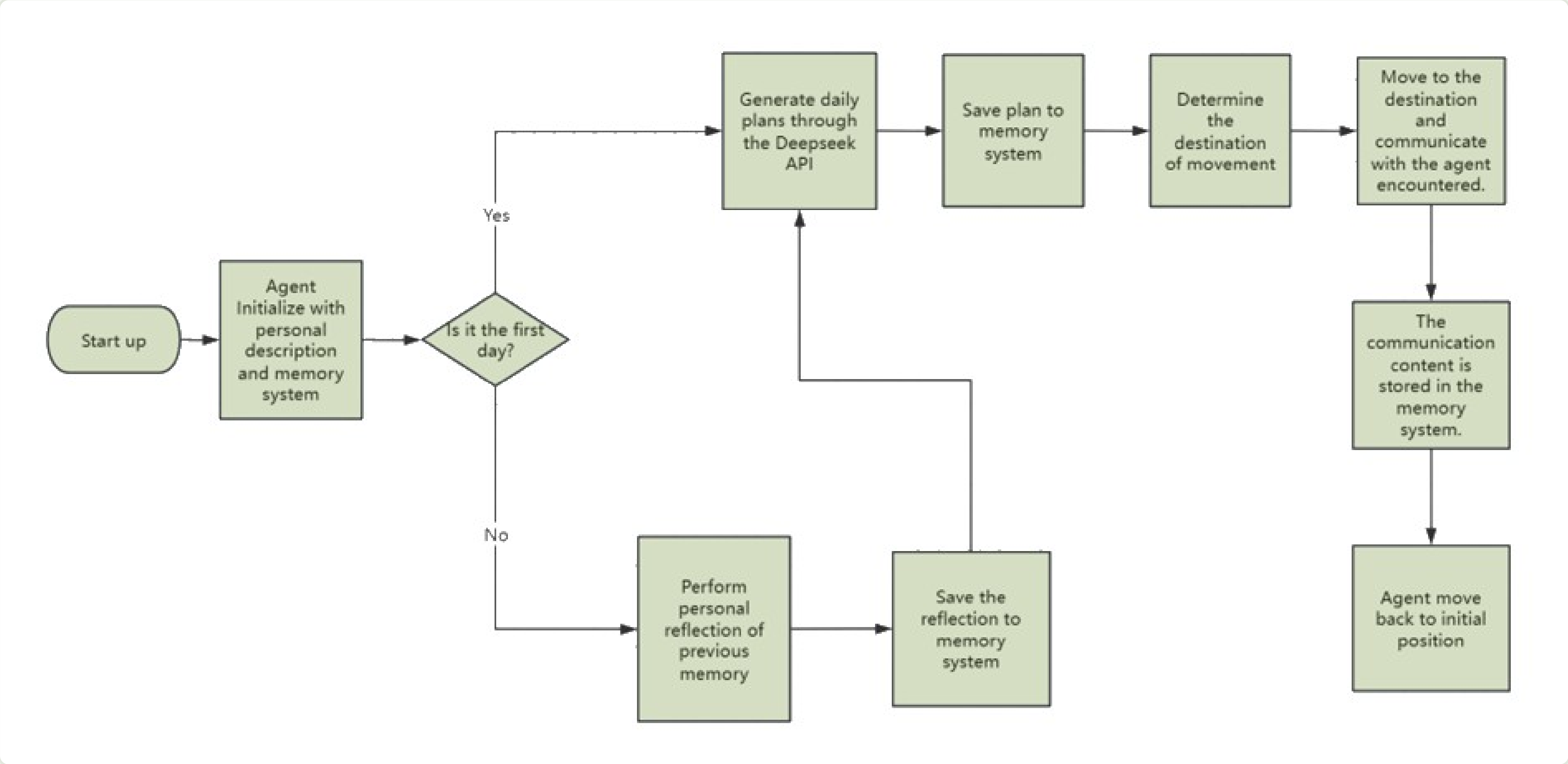
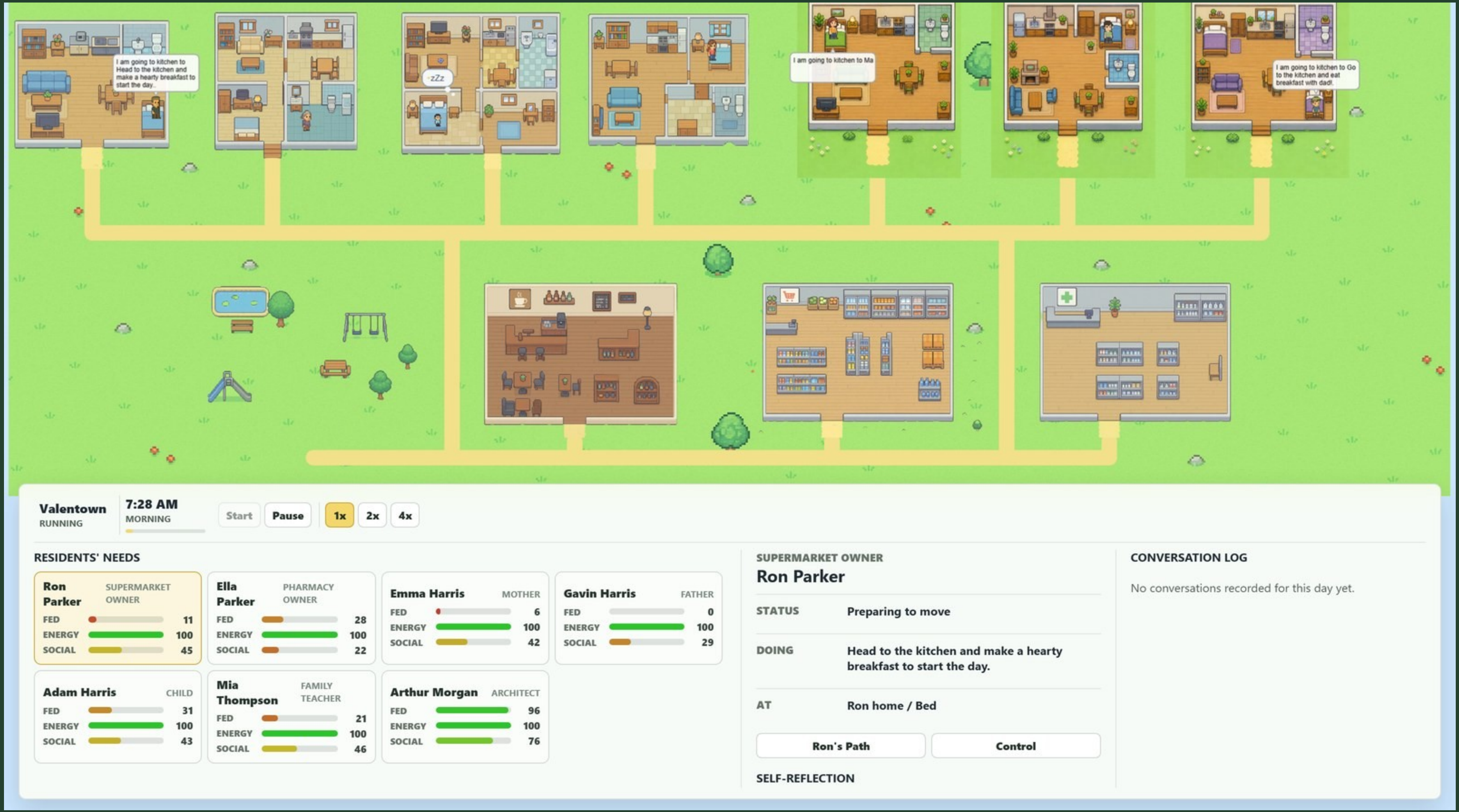


Fig. 2 · Day 1 plans from the profile — every later day reflects first, then plans

Valentown, running



a day planned overnight · two residents meeting at the same place · reflection reshaping tomorrow

EXPERIMENTS

What we ran, and what we looked for

Ron Parker
60 · supermarket owner

Ella Parker
58 · pharmacy owner

Emma Harris
30 · mother

Gavin Harris
32 · father

Adam Harris
7 · child

Mia Thompson
28 · family teacher

Arthur Morgan
29 · architect

Environment

predefined locations;
co-location enables interaction

Duration

multiple consecutive days,
a full cycle each day

Execution

deterministic triggers,
stochastic content

WE LOOK AT

interaction → behaviour

memory → adaptation

emergent social structure

RESULT 1

Turn interaction off, and the town goes quiet

INTERACTION ENABLED

Conversation → Memory → Next day's plan → richer activities · follow-up meetings · lasting pairs

INTERACTION DISABLED

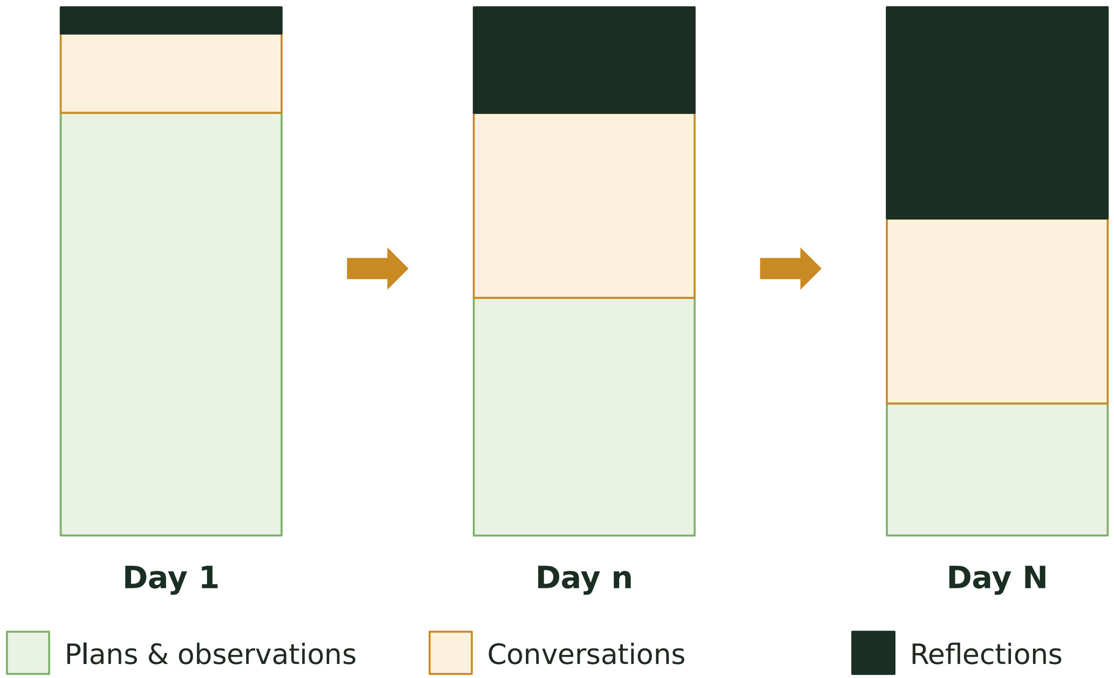
Conversation ✕ Memory → Next day's plan → repetitive · isolated · schedules stay static

Interaction drives behavioural diversity and day-to-day adaptation

RESULT 2

Memory changes shape as the days pass

What memory is made of



schematic — relative composition, not measured proportions

Bonds form on their own

Repeated meetings → persistent pairs → repeated co-location.

Nobody wrote a friendship rule.

Behaviour follows memory

Reflection-derived insight → goal-aligned task selection shifts across days.

Limit memory, and adaptation weakens.

CONCLUSION

Where this leaves us

What we showed

1. Planning, interaction and reflection linked by one memory loop
2. Importance-based memory + reflection → next-day behaviour
3. Consistent routines, lasting bonds, adaptation — no hand-written rules

AND WE OWN THE LIMITS

cost | **scale** | **stochasticity**

quantitative evaluation is the first thing we owe this work

Future work

1 Quantitative evaluation

network metrics, plan coherence, user studies

2 Scale & real time

larger populations, live reaction

3 Deeper reasoning

NPCs, education, embodied agents

Thank you

Questions are very welcome.

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Paper 108

A Lightweight LLM-Driven Multi-Agent Framework for Virtual Society Simulation with Memory Feedback

A backend that thinks, a frontend that shows

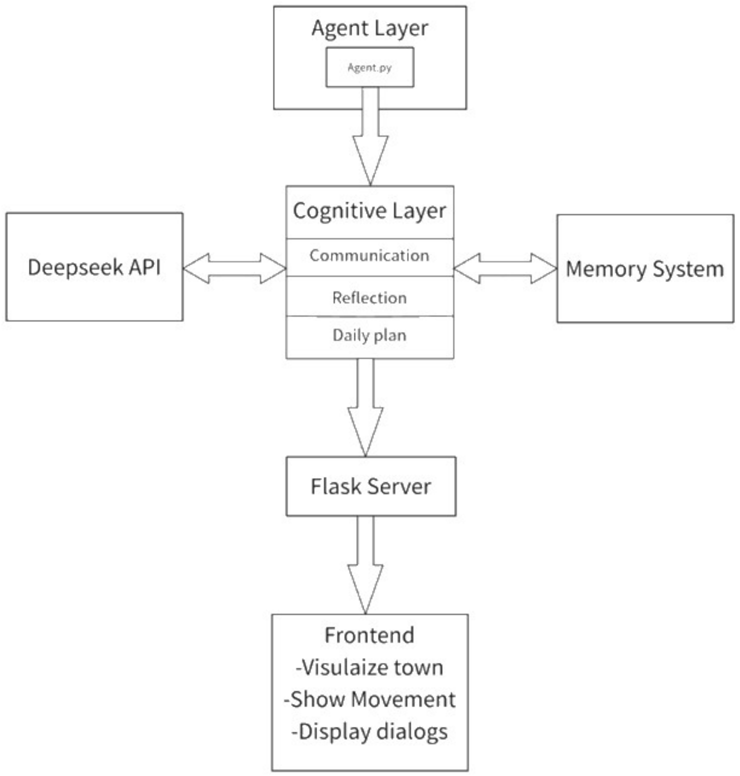

Fig. 1 Overview of the proposed framework

Backend · Python

Planning, dialogue, reflection, memory — the only part that calls the LLM

Frontend · Phaser

Town, movement and conversations, served over a REST API

Five modules

Agent Manager · Scheduler · Interaction Engine · LLM Interface · Memory Manager

Valentown: the framework, running

Valentown

homes · shop · pharmacy · café · park

7 residents

own age, role, personality, goals

DeepSeek API

schedules · dialogue · reflection

Python + Phaser

pre-computed, replayed over REST

BACKUP · DISCUSSION

What this buys us, and what it costs

Advantages

1. **Lightweight & reproducible**
   explicit pipeline, deterministic triggers

2. **Controllable**
   schedules, co-location rules, human feedback

3. **Extendable**
   every module swaps independently

any module can be swapped without touching the others

Limitations

1. **Cost**
   API calls scale with agents × days

2. **Scale**
   small population, simple town

3. **Stochasticity**
   runs vary — strict metrics are hard

agents × **days** × **conversations**

every plan, conversation and reflection is an API call